

Notes: Barrot & Sauvagnat (QJE, 2016)

Input Specificity and the Propagation of Idiosyncratic Shocks in Production Networks

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1 Big picture

- **Question.** Do firm-level idiosyncratic shocks propagate through production networks, or are they absorbed quickly by firms that can substitute across suppliers?
- **Empirical challenge.** Firm-specific shocks are hard to identify because firm performance is jointly determined by demand, sectoral shocks, local shocks, and managerial decisions. The paper solves this by using natural disasters as plausibly exogenous shocks to supplier firms.
- **Core setting.** The authors combine major U.S. natural disasters with supplier-customer links from Compustat Segment data. They follow publicly listed U.S. firms from 1978 to 2013 and observe whether one of a firm’s major suppliers is hit by a disaster.
- **Main downstream result.** When a firm has at least one supplier hit by a major natural disaster, its sales growth drops by roughly 2 to 3 percentage points. Cost of goods sold also falls, consistent with a real input disruption rather than only an accounting or demand effect.
- **Main mechanism.** The propagation is much stronger when the disrupted supplier produces specific inputs. The paper measures specificity using the Rauch classification of differentiated goods, suppliers’ R&D, and suppliers’ patents.
- **Value effect.** Customer firms experience negative cumulative abnormal returns when a supplier is hit. The stock market therefore treats the lost sales as value destroying rather than merely delayed.
- **Horizontal propagation.** A disruption to one supplier can also reduce sales of other suppliers serving the same customer, because the common customer cuts production and buys less from all related suppliers.
- **Identification message.** The results are not driven by local or industry-wide disaster effects. The paper controls for firm fixed effects, year-quarter fixed effects, state-year fixed effects, industry-year fixed effects, and firm-level trends, and it uses inactive supplier-customer links as placebo comparisons.
- **Economic message.** Production networks can transmit firm-level shocks when inputs are difficult to substitute. Input specificity is therefore a key microeconomic reason why idiosyncratic shocks can become aggregate-relevant.

2 Incremental contribution of the paper

2.1 Contribution relative to macro-network theory

The paper provides direct firm-level evidence for a central premise in macro-network models: idiosyncratic shocks can propagate through input-output linkages. Earlier work showed theoretically that sectoral or firm-level shocks may matter for aggregate fluctuations when the production network is granular or has important linkages. Barrot and Sauvagnat move from theory to micro evidence by showing that a shock to one supplier is followed by lower production at its customers.

Interpretation: The contribution is not merely that suppliers and customers are correlated. The authors identify sharp, plausibly exogenous disruptions and follow their transmission through observed firm-to-firm links.

2.2 Contribution relative to firm and trade literatures on input specificity

The paper connects the propagation of shocks to input specificity. The idea that differentiated inputs require relationship-specific investments is common in trade, organization, and incomplete-contracts theory. The paper shows that this friction has macro and asset-pricing consequences: when inputs are harder to replace, temporary supplier disruptions create larger output and market-value losses.

Interpretation: Input specificity is a micro foundation for shock amplification. The more customized the input, the less the customer can substitute away from the shocked supplier.

2.3 Contribution relative to disaster/event identification

The paper uses natural disasters as idiosyncratic supply shocks at the firm level. This is valuable because most firm-level shocks are endogenous or confounded by demand shocks. Disasters are geographically localized, sudden, and plausibly unrelated to the characteristics of a firm’s customers.

Interpretation: This gives the paper a credible shock source that can be mapped through supplier-customer links while controlling for local and industry confounds.

2.4 Contribution relative to asset pricing and corporate finance

The paper shows that input disruptions are capitalized into stock prices. Customer firms lose market value when suppliers are hit, and the stock-price response is larger when the supplier is specific. This links operational supply-chain fragility to shareholder value.

Interpretation: The paper therefore bridges real production networks and financial market valuation. It shows that investors recognize supplier risk as a firm-value risk.

2.5 Contribution relative to empirical production networks

The paper documents not only direct supplier-customer propagation, but also horizontal propagation to other suppliers of the same customer. This shows that the network effect is not confined to one edge. A disruption to one input can reduce the customer’s output and thereby affect other upstream firms.

Interpretation: This expands the empirical view of propagation from pairwise relationships to network-wide production complementarities.

3 Data and measurement

3.1 Natural disasters as firm-level shocks

- The paper identifies shocks using major natural disasters in the United States from the SHELDUS database.
- Disasters are measured at the county level and include hurricanes, earthquakes, floods, storms, wildfires, and similar events.
- A firm is treated when its headquarters county is hit by a major disaster.
- The main treatment indicator is denoted conceptually as $HitFirm_{i,t}$, equal to one if firm i is located in a county hit by a major disaster in quarter t .
- For customer firms, $HitSupplier_{i,t}$ equals one if at least one of firm i ’s suppliers is hit by a major disaster in quarter t .

Interpretation: The disaster design isolates shocks to suppliers that are unlikely to be caused by customers’ demand, suppliers’ managerial choices, or industry-wide trends.

3.2 Supplier-customer network

- Supplier-customer links come from Compustat Segment data.
- Public firms disclose customers that account for a large fraction of sales, so the observed links are economically material.

- The network contains active supplier-customer relationships and allows the authors to compare active links with relationships that are not active in a given year.
- Figure I illustrates how the supplier-customer network changes between 1995 and 2000.

Interpretation: The data do not capture every supplier of every firm. They capture major supply-chain links, which is precisely where disruptions should matter most.

3.3 Main samples

The paper uses two main panel samples.

- The **customer sample** contains firm-quarter observations for firms that are customers in the Compustat Segment network.
- The **supplier sample** contains firm-quarter observations for firms that appear as suppliers.
- Table II reports 80,574 observations in the customer sample and 139,976 observations in the supplier sample.

Interpretation: The split between customer and supplier samples lets the authors estimate both the direct effect of disasters on affected suppliers and the downstream effect on their customers.

3.4 Outcome variables

The two main real outcomes are quarterly growth in sales and cost of goods sold:

$$\Delta Sales_{i,t-4,t} = \frac{Sales_{i,t} - Sales_{i,t-4}}{Sales_{i,t-4}}, \quad (1)$$

$$\Delta COGS_{i,t-4,t} = \frac{COGS_{i,t} - COGS_{i,t-4}}{COGS_{i,t-4}}. \quad (2)$$

- The comparison is to the same quarter in the previous year, which absorbs seasonality.
- Sales growth captures output disruption.
- COGS growth helps distinguish real production effects from pure revenue timing.

Interpretation: If customers cannot obtain critical inputs, both sales and the cost of goods sold should fall. This joint pattern is difficult to reconcile with only demand-side effects.

3.5 Input specificity measures

The paper constructs three measures of supplier specificity:

1. **Differentiated goods.** Suppliers in industries classified by Rauch as differentiated are treated as producing more specific inputs.
2. **R&D intensity.** Suppliers with higher R&D expenses are treated as producing more relationship-specific or customized inputs.
3. **Patent intensity.** Suppliers with more patents are treated as producing inputs with fewer easy substitutes.

Interpretation: All three measures aim to capture the same economic friction: customers cannot quickly replace a supplier when the input is specialized, customized, or technologically protected.

3.6 Stock market outcomes

The paper estimates abnormal returns around supplier disasters.

- It computes cumulative average abnormal returns for customers whose suppliers are hit.
- Abnormal returns are estimated using a three-factor Fama-French model.
- The event window runs around the first day of the natural disaster affecting a supplier.

Interpretation: Market-value losses show that supplier shocks are not just temporary accounting fluctuations. Investors view them as economically meaningful cash-flow losses.

4 Empirical specifications

4.1 Direct effect of disasters on suppliers

The first step estimates how disasters affect the firms that are directly hit. A simplified event-time specification is:

$$\Delta Sales_{i,t-4,t} = \alpha + \sum_{\tau=-4}^9 \beta_{\tau} HitFirm_{i,t-\tau} + \eta_i + \pi_t + \epsilon_{i,t}. \quad (3)$$

- η_i are firm fixed effects.
- π_t are year-quarter fixed effects.
- β_{τ} traces the dynamics before and after the disaster.

Interpretation: The direct effect is important because downstream propagation can only be interpreted as supply-chain transmission if disasters actually reduce the output of the affected suppliers.

4.2 Baseline downstream propagation

The core downstream regression is:

$$\Delta Sales_{i,t-4,t} = \alpha + \beta_0 HitSupplier_{i,t} + \beta_4 HitSupplier_{i,t-4} + \Gamma X_{i,t} + \eta_i + \pi_t + \epsilon_{i,t}. \quad (4)$$

- $HitSupplier_{i,t}$ captures contemporaneous exposure to a supplier hit by a disaster.
- $HitSupplier_{i,t-4}$ captures exposure in the same quarter of the previous year.
- $X_{i,t}$ includes controls such as the number of suppliers and interactions of size, age, and ROA with time effects.
- Additional specifications include state-year and industry-year fixed effects.

Interpretation: A negative β_4 means that a shock to a supplier reduces the customer's subsequent annual sales growth.

4.3 Dynamic downstream propagation

The dynamic version includes leads and lags:

$$\Delta Sales_{i,t-4,t} = \alpha + \sum_{\tau=-5}^1 \gamma_{\tau} HitSupplier_{i,t-\tau} + \sum_{\tau=-5}^1 \delta_{\tau} HitFirm_{i,t-\tau} + \eta_i + \pi_t + \epsilon_{i,t}. \quad (5)$$

Interpretation: This tests whether the effect is concentrated after the supplier shock and whether customers themselves are directly hit by disasters.

4.4 Input specificity interactions

To test the mechanism, the paper estimates:

$$\begin{aligned} \Delta Sales_{i,t-4,t} = & \alpha + \beta_1 HitSpecificSupplier_{i,t-4} + \beta_2 HitNonspecificSupplier_{i,t-4} \\ & + \beta_3 HitFirm_{i,t-4} + \eta_i + \pi_t + \Gamma X_{i,t} + \epsilon_{i,t}. \end{aligned} \quad (6)$$

Interpretation: The model predicts $|\beta_1| > |\beta_2|$: specific suppliers should generate stronger downstream losses because customers face higher switching costs.

4.5 Market value effect

The event-study specification uses abnormal returns:

$$CAAR_{g,[a,b]} = \frac{1}{N_g} \sum_{i \in g} \sum_{d=a}^b \widehat{AR}_{i,d}, \quad (7)$$

where g indexes customers, suppliers, or control firms.

Interpretation: If the lost sales are simply delayed, the stock market response should be small. A negative customer CAAR indicates that input disruptions destroy firm value.

4.6 Horizontal propagation

The horizontal propagation tests ask whether a supplier of a customer is affected when another supplier of that same customer is hit:

$$\Delta Sales_{j,t-4,t} = \alpha + \beta HitCustomerSupplier_{j,t-4} + \eta_j + \pi_t + \epsilon_{j,t}. \quad (8)$$

Interpretation: A negative coefficient means the shock travels from one supplier to the customer and then to the customer's other suppliers.

5 Identification and empirical design

5.1 Why natural disasters identify idiosyncratic shocks

- Natural disasters are sudden, plausibly exogenous, and geographically localized.
- They affect firms because of physical location rather than demand or managerial behavior.
- The paper uses county-level exposure, which maps disasters to firm headquarters.
- Disasters that affect many firms or large industries are checked separately to ensure the results are not sector-wide shocks.

Interpretation: The identification assumption is that, conditional on fixed effects and controls, supplier disasters affect customers primarily through the supplier-customer relationship.

5.2 Ruling out common local and industry shocks

The paper uses several strategies:

- State-year fixed effects absorb local economic shocks.
- Industry-year fixed effects absorb industry demand and supply shocks.

- Firm fixed effects absorb time-invariant firm characteristics.
- Year-quarter fixed effects absorb macro shocks.
- Inactive supplier-customer relationships act as placebo links.

Interpretation: These tests address the concern that customer losses are caused by the same disaster directly affecting the customer, by local demand disruption, or by industry-wide shocks.

5.3 Why input specificity sharpens the causal interpretation

- If the estimates were driven by broad local demand shocks, they should not systematically vary with whether the supplier produces differentiated, R&D-intensive, or patented inputs.
- The paper finds stronger effects for specific suppliers across all three measures.
- This pattern matches the switching-cost mechanism and is difficult to explain with unrelated local shocks.

Interpretation: Heterogeneity by specificity is the main mechanism test. It shows that the propagation effect is not just a generic supplier exposure effect.

5.4 Threats and limitations

- Compustat Segment data observe only major disclosed customer relationships.
- The sample is restricted to publicly listed firms.
- Firm headquarters may not perfectly capture the physical location of production facilities.
- Disaster intensity is measured at the county level and cannot capture all plant-level disruption.
- Network links may be endogenous, although the disaster timing is plausibly exogenous.

Interpretation: These limitations mostly affect external validity and measurement precision. The extensive robustness tests help address internal validity.

6 Tables and figures

6.1 Figure I and Table I: network and disaster shocks

Read:

- Figure I illustrates the supplier-customer network and how relationships change from 1995 to 2000.
- Table I lists the major natural disasters used as shocks, their dates, the number of counties affected, employment exposure, and locations.
- Disasters vary widely in geographic scope and employment exposure, which creates variation in firm-level treatment.

Economic message: The network figure shows the potential channels for propagation, while Table I shows the exogenous shock events used to activate those channels.

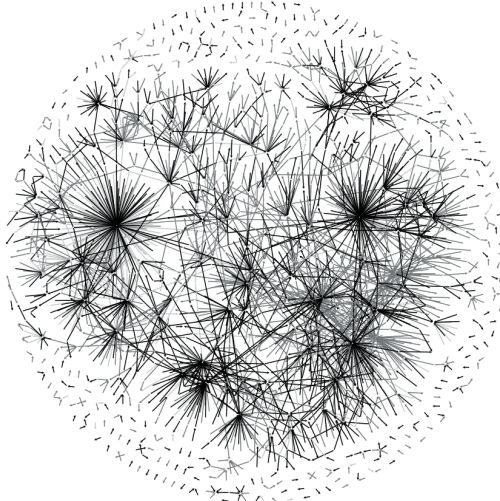


FIGURE I
Network Evolution from 1995 to 2000

(a) Figure I: supplier-customer network evolution

Disaster	Date	# Counties	U.S. Employment Affected (%)	Location
Mount St. Helens eruption	May 1980	2	0.03	WA
Hurricane Alicia	August 1983	139	4.72	TX
Hurricane Elena	August 1985	32	0.54	AL, FL, LA, MS
Hurricane Juan	October 1985	66	3.58	AL, FL, LA, MS, TX
Hurricane Hugo	September 1989	71	1.43	NC, SC, VA
Loma earthquake	October 1989	8	2.56	CA
Hurricane Bob	August 1991	54	7.06	MA, ME, NC, NH, NY, RI
Oakland Hills firestorm	October 1991	1	0.54	CA
Hurricane Andrew	August 1992	51	2.67	AL, FL, LA, MS
Hurricane Iniki	September 1992	1	0.02	HI
Blizzard	March 1993	221	11.15	AL, CT, FL, GA, MA, MD, NJ, OH, SC, VA, VT
Northridge earthquake	January 1994	1	3.69	CA
Hurricane Alberto	July 1994	41	0.66	AL, FL, GA
Hurricane Opal	October 1995	186	6.43	AL, FL, GA, LA, MS, NC, SC
Blizzard	January 1996	319	14.57	CT, DE, IN, KY, MA, MD, NC, NJ, NY, PA, VA, WV
Hurricane Fran	September 1996	190	2.02	NC, SC, VA, WV
Ice storm	January 1998	43	1.09	ME, NH, NY, VT
Hurricane Bonnie	August 1998	43	1.26	NC, VA
Hurricane Georges	September 1998	78	3.68	AL, FL, LA, MS
Hurricane Floyd	September 1999	226	15.68	CT, DC, DE, FL, MD, ME, NC, NH, NJ, NY, PA, SC, VA, VT
Hurricane Allison	June 2001	77	4.56	AL, FL, GA, LA, MS, PA, TX
Hurricane Isabel	September 2003	89	4.99	DE, MD, NC, NJ, NY, PA, RI, VA, VT, WV

(b) Table I: major disasters, part 1

Disaster	Date	# Counties	U.S. Employment Affected (%)	Location
Southern California wildfires	October 2003	3	1.78	CA
Hurricane Charley	August 2004	67	3.94	FL, GA, NC, SC
Hurricane Frances	September 2004	311	12.47	AL, FL, GA, KY, MD, NC, NY, OH, PA, SC, VA, WV
Hurricane Ivan	September 2004	284	7.31	AL, FL, GA, KY, LA, MA, MD, MS, NC, NH, NJ, NY, PA, SC, TN, WV
Hurricane Jeanne	September 2004	160	8.8	DE, FL, GA, MD, NC, NJ, PA, SC, VA
Hurricane Dennis	July 2005	200	5.38	AL, FL, GA, MS, NC
Hurricane Katrina	August 2005	288	9.21	AL, AR, FL, GA, IN, KY, LA, MI, MS, OH, TN
Hurricane Rita	September 2005	123	3.75	AL, AR, FL, LA, MS
Hurricane Wilma	October 2005	24	3.55	FL
Midwest floods	June 2008	216	5.25	IA, IL, IN, MN, MO, NE, WI
Hurricane Gustav	September 2008	98	1.79	AR, LA, MS
Hurricane Ike	September 2008	163	4.11	AR, LA, MO, TN, TX
Blizzard, Groundhog Day	February 2011	210	14.63	CT, IA, IL, IN, KS, MA, MO, NJ, NM, NY, OH, OK, PA, TX, WI
Hurricane Irene	August 2011	40	3.19	CT, MA, MD, NC, NJ, NY, VA, VT
Tropical Storm Lee	September 2011	110	5.23	AL, CT, GA, LA, MD, MS, NJ, NY, PA, TN, VA
Isaac	August 2012	77	3.36	FL, LA, MS
Hurricane Sandy	October 2012	274	22.08	CT, DE, MA, MD, NC, NH, NJ, NY, OH, PA, RI, VA, WV
Flooding and severe weather, Illinois	April 2013	29	3.11	IL, IN, MO
Flooding, Colorado	September 2013	8	0.92	CO

Notes. This table describes the 41 natural disasters included in the sample. Names, dates, number of affected counties, and the location of each natural disaster are obtained from the SHELUS database at the University of South Carolina. The list is restricted to events classified as Major Disasters in SHELUS, with total direct estimated damages above \$1 billion 2013 constant dollars and lasting less than 30 days. The share of total U.S. employment affected by each natural disaster is computed from County Business Pattern data publicly provided by the U.S. Census Bureau. The sample period is from January 1978 to December 2013.

6.2 Figures II and III: geography of disasters and suppliers

Read:

- Figure II maps the frequency of major natural disasters by U.S. county.
- Figure III maps the locations of suppliers' headquarters.
- The maps show that both disaster exposure and supplier locations vary substantially across space.

Economic message: Spatial variation is central to identification. Supplier locations determine which network links are shocked by a disaster.

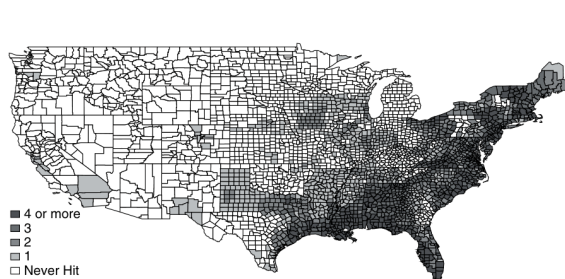


FIGURE II

Major Natural Disaster Frequency by U.S. Counties

(a) Figure II: disaster frequency by county

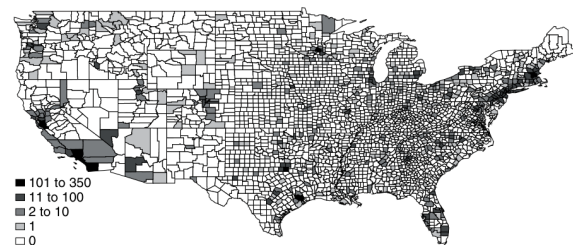


FIGURE III

Location of Sample Suppliers' Headquarters

This map presents for our sample the number of suppliers' headquarters located in each U.S. county. Data on the location of headquarters are obtained

(b) Figure III: supplier headquarters

6.3 Table II: descriptive statistics

Read:

- The customer sample has 80,574 firm-quarter observations; the supplier sample has 139,976 observations.
- The mean customer sales growth is about 10.2% and the mean COGS growth is about 10.6%.
- Customers have an average of 1.38 major suppliers in the sample.
- Supplier-customer relationships last about seven years on average.
- The average disclosed supplier input share is about 2.2% of the customer's cost of goods sold.

Economic message: The disclosed suppliers are only a small share of total inputs, so the estimated customer sales losses are economically large.

TABLE II
DESCRIPTIVE STATISTICS

	Obs.	Mean	Std. Dev.	p1	p50	p99
Panel A: Customer sample						
Sales growth ($t - 4, t$)	80,574	0.102	0.375	-0.606	0.040	1.927
Cogs growth ($t - 4, t$)	79358	0.106	0.411	-0.651	0.038	2.193
Disaster hits firm (t)	80,574	0.016	0.126	0.000	0.000	1.000
Disaster hits one supplier (t)	80,574	0.014	0.118	0.000	0.000	1.000
Number of suppliers	80,574	1.383	4.162	0.000	0.000	19.000
	Diff.		R&D		Patent	
	S	NS	S	NS	S	NS
Av. duration of relationships	7.125	6.692	6.373	8.335	7.821	6.618
Av. supplier-customer HQs distance	1,332	1,210	1,502	1,214	1,388	1,219
Av. suppliers' input share	0.022	0.025	0.017	0.023	0.025	0.022
	Eventually treated			Never treated		
	Obs.	Mean	Std. dev.	Obs.	Mean	Std. dev.
Assets	32,061	12,656	20,013	48,513	3,254	7,099
Age	32,061	27.822	16.623	48,513	19.233	15.680
ROA	32,061	0.145	0.091	48,513	0.118	0.128
Panel B: Supplier sample						
	Obs.	Mean	Std. dev.	p1	p50	p99
Sales growth ($t - 4, t$)	139,976	0.188	0.814	-0.876	0.045	4.568
Disaster hits firm (t)	139,976	0.017	0.127	0.000	0.000	1.000
Disaster hits a customer (t)	139,976	0.008	0.088	0.000	0.000	0.000
Disaster hits a customer's supplier (t)	139,976	0.042	0.200	0.000	0.000	1.000
Number of customers	139,976	0.711	0.964	0.000	0.000	4.000
% Employees at HQs	102,279	0.597	0.365	0.000	0.667	1.000

6.4 Tables III and IV: direct supplier effects and input specificity

Read:

- Table III shows that disasters reduce sales growth of directly hit suppliers for several quarters after the event.
- The direct supplier effect is negative and statistically significant, confirming that the disasters are real supply shocks.
- Table IV shows that the direct supplier effect is present across specificity measures.
- The estimates help rule out the possibility that specific suppliers are mechanically hit by more severe disasters.

Economic message: Before studying propagation, the paper first confirms that disasters lower output at affected suppliers. This direct effect is the source shock.

TABLE III
NATURAL DISASTER DISRUPTIONS—SUPPLIER SALES GROWTH

	Sales Growth ($t - 4, t$)			
Disaster hits firm (t)	-0.006 (0.018)	-0.004 (0.018)	-0.001 (0.018)	-0.011 (0.018)
Disaster hits firm ($t - 1$)	-0.045*** (0.016)	-0.045*** (0.016)	-0.032* (0.017)	-0.039** (0.018)
Disaster hits firm ($t - 2$)	-0.033* (0.018)	-0.032* (0.018)	-0.024 (0.021)	-0.026 (0.021)
Disaster hits firm ($t - 3$)	-0.042** (0.019)	-0.040** (0.019)	-0.032 (0.022)	-0.029 (0.023)
Disaster hits firm ($t - 4$)	-0.031 (0.020)	-0.028 (0.020)	-0.029 (0.022)	-0.024 (0.023)
Disaster hits firm ($t - 5$)	-0.007 (0.020)	-0.005 (0.020)	-0.022 (0.023)	-0.019 (0.023)
Firm FE	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes
Size, age, ROA $\times$ year-quarter FE	No	Yes	Yes	Yes
State-year FE	No	No	Yes	Yes
Industry-year FE	No	No	No	Yes
Observations	139,976	139,976	139,976	139,976
R^2	0.177	0.192	0.212	0.233

Notes. This table presents estimates from panel regressions of firms' sales growth relative to the same

(a) Table III: direct supplier sales growth effect

TABLE IV
NATURAL DISASTERS DISRUPTIONS—SPECIFIC VERSUS NONSPECIFIC SUPPLIERS

	Sales Growth ($t - 4, t$)					
Supplier specificity:	Diff.		R&D		Patent	
Disaster hits firm ($t - 4, t - 1$)	-0.050*** (0.017)	-0.044*** (0.016)	-0.048*** (0.012)	-0.048*** (0.012)	-0.046*** (0.016)	-0.041*** (0.015)
Disaster hits specific firm ($t - 4, t - 1$)	0.023 (0.026)	0.013 (0.026)	0.038 (0.040)	0.044 (0.039)	0.020 (0.028)	0.011 (0.028)
Specific firm			0.099*** (0.021)	0.090*** (0.021)	-0.060*** (0.014)	-0.030** (0.013)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Size, age, ROA $\times$ year-quarter FE	No	Yes	No	Yes	No	Yes
Observations	139,976	139,976	139,976	139,976	139,976	139,976
R^2	0.177	0.192	0.177	0.192	0.177	0.192

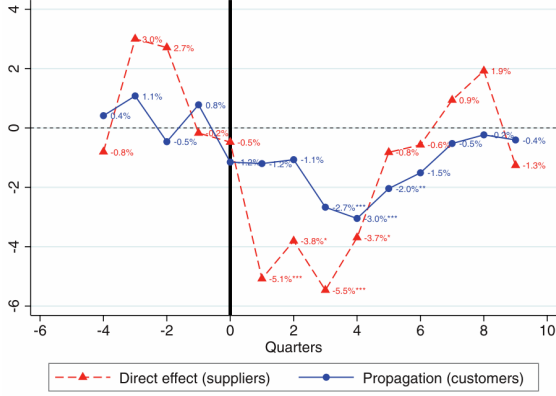
(b) Table IV: direct effect by supplier specificity

6.5 Figure IV and Table V: downstream propagation baseline

Read:

- Figure IV plots sales growth around disaster events for directly hit suppliers and their customers.
- Direct suppliers suffer large sales declines after the event, and customers also experience negative sales growth after supplier disruptions.
- Table V estimates the baseline downstream effect on sales and COGS.
- A supplier disaster reduces customer sales growth by about 2–3 percentage points and reduces COGS growth as well.

Economic message: Supplier disruptions generate real production losses for customers, not merely changes in revenue timing or financial reporting.



(a) Figure IV: natural disasters and sales growth

TABLE V
DOWNSTREAM PROPAGATION—BASELINE

Panel A		Sales Growth ($t - 4, t$)			
Disaster hits one supplier ($t - 4$)	-0.031*** (0.009)	-0.027*** (0.008)	-0.029*** (0.008)	-0.019** (0.008)	
Disaster hits firm ($t - 4$)	-0.031*** (0.011)	-0.029*** (0.011)	-0.005 (0.009)	-0.003 (0.009)	
Number of suppliers	Yes	Yes	Yes	Yes	
Firm FE	Yes	Yes	Yes	Yes	
Year-quarter FE	Yes	Yes	Yes	Yes	
Size, age, ROA $\times$ year-quarter FE	No	Yes	Yes	Yes	
State-year FE	No	No	Yes	Yes	
Industry-year FE	No	No	No	Yes	
Observations	80,574	80,574	80,574	80,574	
R^2	0.234	0.262	0.300	0.342	
Panel B		Cost of Goods Sold Growth ($t - 4, t$)			
Disaster hits one supplier ($t - 4$)	-0.031*** (0.010)	-0.028*** (0.010)	-0.029*** (0.010)	-0.020** (0.010)	
Disaster hits firm ($t - 4$)	-0.014 (0.012)	-0.013 (0.012)	0.001 (0.011)	0.002 (0.011)	
Number of suppliers	Yes	Yes	Yes	Yes	
Firm FE	Yes	Yes	Yes	Yes	
Year-quarter FE	Yes	Yes	Yes	Yes	
Size, age, ROA $\times$ year-quarter FE	No	Yes	Yes	Yes	
State-year FE	No	No	Yes	Yes	
Industry-year FE	No	No	No	Yes	
Observations	79,358	79,358	79,358	79,358	
R^2	0.188	0.215	0.253	0.290	

(b) Table V: baseline downstream propagation

6.6 Tables VI and VII: dynamics and robustness

Read:

- Table VI estimates dynamic effects with multiple leads and lags.
- The strongest customer losses appear after the supplier shock, especially around $t - 3$ and $t - 4$ relative to the sales-growth window.
- Table VII shows that results are robust to controlling for the share of the workforce directly affected and for eventually linked suppliers.
- These checks reduce concerns that the effect is caused by local disaster exposure or common geography rather than an active supplier link.

Economic message: The timing and placebo-link evidence support the interpretation that shocks propagate through active input relationships.

TABLE VI
DOWNSTREAM PROPAGATION—SALES GROWTH DYNAMICS

	Sales Growth ($t - 4, t$)			
Disaster hits one supplier (t)	-0.012 (0.008)	-0.010 (0.008)	-0.007 (0.008)	-0.003 (0.008)
Disaster hits one supplier ($t - 1$)	-0.013 (0.008)	-0.013 (0.009)	-0.011 (0.009)	-0.004 (0.009)
Disaster hits one supplier ($t - 2$)	-0.013 (0.009)	-0.009 (0.009)	-0.010 (0.010)	0.002 (0.010)
Disaster hits one supplier ($t - 3$)	-0.028*** (0.009)	-0.025*** (0.009)	-0.025*** (0.009)	-0.013 (0.009)
Disaster hits one supplier ($t - 4$)	-0.031*** (0.009)	-0.027*** (0.009)	-0.030*** (0.009)	-0.020** (0.009)
Disaster hits one supplier ($t - 5$)	-0.016* (0.010)	-0.013 (0.010)	-0.014 (0.010)	-0.007 (0.010)
Disaster hits firm (t)	0.015 (0.012)	0.016 (0.012)	0.015 (0.012)	0.011 (0.012)
Disaster hits firm ($t - 1$)	-0.003 (0.011)	-0.003 (0.011)	0.001 (0.011)	-0.003 (0.012)
Disaster hits firm ($t - 2$)	-0.023** (0.011)	-0.022** (0.011)	-0.002 (0.013)	0.002 (0.013)
Disaster hits firm ($t - 3$)	-0.042*** (0.011)	-0.043*** (0.011)	-0.022* (0.013)	-0.016 (0.013)
Disaster hits firm ($t - 4$)	-0.034*** (0.012)	-0.032*** (0.012)	-0.010 (0.013)	-0.006 (0.013)
Disaster hits firm ($t - 5$)	-0.026** (0.012)	-0.027** (0.012)	-0.010 (0.012)	-0.006 (0.012)
Number of suppliers	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes
Size, age, ROA $\times$ year-quarter FE	No	Yes	Yes	Yes
State-year FE	No	No	Yes	Yes
Industry-year FE	No	No	No	Yes
Observations	80,574	80,574	80,574	80,574
R^2	0.234	0.262	0.300	0.342

(a) Table VI: downstream sales growth dynamics

TABLE VII
DOWNSTREAM PROPAGATION—ROBUSTNESS

	Sales Growth ($t - 4, t$)			
Panel A: Controlling for share of the workforce hit				
Disaster hits more than 10% of firm's workforce ($t - 4$)	-0.006 (0.010)	-0.005 (0.010)	0.002 (0.010)	0.009 (0.010)
Disaster hits one supplier ($t - 4$)	-0.031*** (0.009)	-0.027*** (0.009)	-0.030*** (0.009)	-0.020** (0.008)
Disaster hits firm ($t - 4$)	-0.027** (0.012)	-0.026** (0.012)	-0.006 (0.011)	-0.009 (0.011)
Number of suppliers	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes
Size, age, ROA $\times$ year-quarter FE	No	Yes	Yes	Yes
State-year FE	No	No	Yes	Yes
Industry-year FE	No	No	No	Yes
Observations	80,574	80,574	80,574	80,574
R^2	0.234	0.262	0.300	0.342
Panel B: Controlling for whether any "eventually linked" supplier is hit				
Disaster hits <i>any eventually</i> <i>linked suppliers'</i> location ($t - 4$)	0.003 (0.007)	0.004 (0.007)	0.004 (0.007)	0.005 (0.007)
Disaster hits one supplier ($t - 4$)	-0.033*** (0.010)	-0.029*** (0.010)	-0.032*** (0.010)	-0.023** (0.010)
Disaster hits firm ($t - 4$)	-0.031*** (0.011)	-0.029*** (0.011)	-0.005 (0.009)	-0.003 (0.009)
Number of suppliers	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes
Size, age, ROA $\times$ year-quarter FE	No	Yes	Yes	Yes
State-year FE	No	No	Yes	Yes
Industry-year FE	No	No	No	Yes
Observations	80,574	80,574	80,574	80,574
R^2	0.234	0.262	0.300	0.342

(b) Table VII: robustness checks

6.7 Table VIII and Table IX: input specificity and market value

Read:

- Table VIII shows that downstream propagation is concentrated among specific suppliers.
- Across the differentiated-goods, R&D, and patent measures, specific suppliers generate large negative customer sales effects.
- Nonspecific suppliers produce smaller and often insignificant effects.
- Table IX shows that customers lose market value when a supplier is hit by a disaster.
- Customers lose about 1.1% over the full event window, while directly hit suppliers lose more.

Economic message: Specificity explains why shocks propagate, and stock prices show that the resulting output losses are value relevant.

TABLE VIII
DOWNSTREAM PROPAGATION—INPUT SPECIFICITY

Supplier Specificity	Sales Growth ($t - 4, t$)					
	Diff.		R&D		Patent	
Disaster hits one nonspecific supplier ($t - 4$)	-0.002 (0.012)	-0.002 (0.011)	-0.018 (0.011)	-0.011 (0.011)	-0.020* (0.011)	-0.016 (0.010)
Disaster hits one specific supplier ($t - 4$)	-0.050*** (0.010)	-0.043*** (0.010)	-0.039*** (0.014)	-0.032** (0.014)	-0.039*** (0.011)	-0.034*** (0.012)
Disaster hits firm ($t - 4$)	-0.031*** (0.011)	-0.029*** (0.011)	-0.031*** (0.011)	-0.029*** (0.011)	-0.031*** (0.011)	-0.029*** (0.011)
Number of suppliers	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Size, age, ROA $\times$ year-quarter FE	No	Yes	No	Yes	No	Yes
Observations	80,574	80,574	80,574	80,574	80,574	80,574
R^2	0.234	0.262	0.234	0.261	0.234	0.262

Notes. This table presents estimates from panel regressions of firms' sales growth relative to the same quarter in the previous year on two dummies indicating whether (at least) one specific supplier and whether (at least) one nonspecific supplier is hit by a major disaster in the same quarter of the previous

(a) Table VIII: downstream propagation by input specificity

TABLE IX
DOWNSTREAM PROPAGATION—EFFECT ON FIRM VALUE

	CAAR		
	Customers ($N = 1,082$)	Suppliers (Direct Effect) ($N = 2,004$)	Customers (Control Group) ($N = 6,379$)
$[-10, -1]$	-0.487 (-1.283)	-0.195 (-0.819)	-0.176** (-2.310)
$[0, 10]$	-0.361* (-1.911)	-0.548** (-2.215)	-0.124 (-0.302)
$[11, 20]$	-0.177 (-0.269)	-1.452*** (-3.340)	-0.029 (-0.205)
$[21, 30]$	-0.121 (-0.583)	-0.385 (-1.088)	-0.006 (0.392)
$[31, 40]$	0.014 (-0.215)	0.120 (1.123)	-0.042 (-1.208)
$[-10, 40]$	-1.132** (-1.982)	-2.459*** (-3.029)	-0.379 (-1.563)

Notes. This table presents CAAR of customer firms around the first day of a natural disaster affecting (at least) one of its suppliers. When more than one supplier is affected by the same natural disaster, the event day is the earliest date across affected suppliers reported in SHELDDUS database. Abnormal returns are computed after estimating, for each firm-disaster pair, a three-factor Fama-French model over the interval from 260 to 11 trading days before the event date. We exclude firm-disaster observations with missing returns in the estimation or event windows, when the firm itself is hit by the disaster, or when

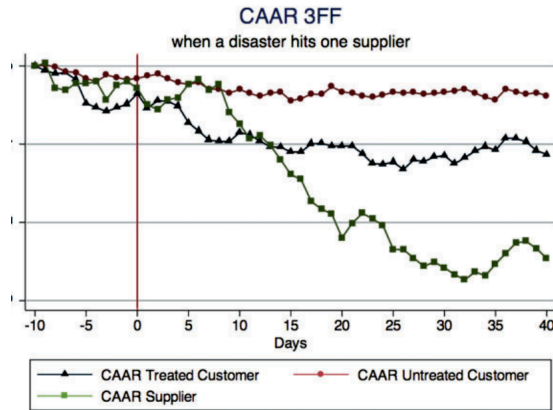
(b) Table IX: customer and supplier CAAR

6.8 Figure V and Table X: event returns and specificity in value losses

Read:

- Figure V plots cumulative abnormal returns around the first day of a supplier disaster.
- Suppliers exhibit the largest negative CAAR, treated customers also decline, and untreated customers remain roughly flat.
- Table X shows that customer CAAR losses are larger when the hit supplier is specific.

Economic message: Equity investors price supplier disruptions, especially when the disrupted input is hard to substitute.



(a) Figure V: cumulative average abnormal returns

TABLE X
DOWNSTREAM PROPAGATION—INPUT SPECIFICITY AND EFFECT ON FIRM VALUE

Supplier Specificity	Customers' CAAR When Disaster Hits at Least One Supplier					
	Diff. $N = 628$	$N = 454$	R&D $N = 318$	$N = 764$	Patent $N = 375$	$N = 707$
At least one specific supplier	Yes	No	Yes	No	Yes	No
$[-10, -1]$	-0.885 (-1.567)	0.064 (-0.288)	-0.321 (-0.568)	-0.556 (-1.243)	-0.096 (0.164)	-0.694* (-1.731)
$[0, 10]$	-0.636*** (-2.757)	0.018 (-0.022)	-0.621 (-1.395)	-0.253 (-1.542)	-0.651*** (-2.585)	-0.208 (-0.742)
$[11, 20]$	-0.168 (-0.608)	-0.189 (0.250)	0.181 (1.482)	-0.326 (-1.021)	0.177 (0.479)	-0.365 (-0.674)
$[21, 30]$	-0.460 (-1.017)	0.348 (0.192)	-1.351** (-2.349)	0.391 (0.564)	-0.560 (-0.502)	0.112 (-0.421)
$[31, 40]$	-0.219 (-0.331)	0.337 (0.006)	-0.340 (-0.896)	0.162 (0.213)	-0.391 (-0.381)	0.229 (-0.038)
$[-10, 40]$	-2.368*** (-2.939)	0.578 (0.080)	-2.452* (-1.769)	-0.582 (-1.412)	-1.519 (-1.285)	-0.926* (-1.690)

Notes. This table presents CAAR of customer firms separately for events affecting (at least) one specific supplier or only nonspecific suppliers. In the first column, a supplier is considered as specific if its industry lies above the median of the share of differentiated goods according to the classification provided by Rauch (1999). In the third column, a supplier is considered specific if the ratio of its R&D expenses over sales is above the median in the two years prior to any given quarter. In the fifth column, a

(b) Table X: CAAR by supplier specificity

6.9 Tables XI and XII: horizontal propagation

Read:

- Table XI shows that when one supplier is hit, other suppliers of the same customer suffer lower sales growth.

- The result is strongest when the shocked supplier is specific.
- Table XII shows robustness of horizontal propagation to controls for workforce exposure, eventually linked customers, and other related suppliers.

Economic message: Shocks do not stop at the first customer. They can propagate horizontally through common customers to other suppliers.

TABLE XI
HORIZONTAL PROPAGATION—RELATED SUPPLIERS' SALES GROWTH

Supplier Specificity	Sales Growth ($t - 4, t$)			
	Diff.	R&D	Patent	
Disaster hits firm ($t - 4, t - 1$)	-0.040*** (0.013)	-0.040*** (0.013)	-0.041*** (0.013)	-0.040*** (0.013)
Disaster hits one customer ($t - 4, t - 1$)	0.002 (0.021)	0.001 (0.021)	0.001 (0.021)	0.002 (0.021)
Disaster hits one customer's supplier ($t - 4, t - 1$)	-0.038*** (0.010)			
Disaster hits one customer's specific supplier ($t - 4, t - 1$)	-0.047*** (0.013)	-0.048*** (0.014)	-0.040*** (0.013)	
Disaster hits one customer's non-specific supplier ($t - 4, t - 1$)	-0.011 (0.013)	-0.013 (0.013)	-0.015 (0.013)	
Number of customers' Suppliers	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes
Size, age, ROA $\times$ year-quarter FE	Yes	Yes	Yes	Yes
Observations	139,976	139,976	139,976	139,976
R^2	0.192	0.192	0.192	0.192

Notes. This table presents estimated coefficients from panel regressions of firms' sales growth relative to the same quarter in the previous year on one dummy indicating whether one of the firm's customers'...

(a) Table XI: related suppliers' sales growth

TABLE XII
HORIZONTAL PROPAGATION—ROBUSTNESS

	Supplier's Sales Growth ($t - 4, t$)			
	Diff.	R&D	Patent	
Panel A: Controlling for share of the workforce hit				
Disaster hits at least 10% of firm's workforce ($t - 4, t - 1$)	-0.007 (0.016)	-0.007 (0.016)	-0.008 (0.016)	-0.007 (0.016)
Disaster hits firm ($t - 4, t - 1$)	-0.035* (0.018)	-0.035** (0.018)	-0.035* (0.018)	-0.035** (0.018)
Disaster hits one customer ($t - 4, t - 1$)	0.002 (0.021)	0.001 (0.021)	0.001 (0.021)	0.002 (0.021)
Disaster hits one customer's supplier ($t - 4, t - 1$)	-0.038*** (0.010)			
Disaster hits one customer's specific supplier ($t - 4, t - 1$)	-0.047*** (0.013)	-0.048*** (0.014)	-0.040*** (0.013)	
Disaster hits one customer's nonspecific supplier ($t - 4, t - 1$)	-0.011 (0.013)	-0.013 (0.013)	-0.015 (0.013)	
Observations	139,976	139,976	139,976	139,976
R^2	0.192	0.192	0.192	0.192
Panel B: Controlling for whether any customers' "eventually linked supplier" is hit				
Disaster hits <i>any customers' eventually linked suppliers'</i> location ($t - 4, t - 1$)	-0.022* (0.013)	-0.022* (0.013)	-0.022 (0.013)	-0.022* (0.013)
Disaster hits firm ($t - 4, t - 1$)	-0.040*** (0.013)	-0.041*** (0.013)	-0.041*** (0.013)	-0.041*** (0.013)
Disaster hits one customer ($t - 4, t - 1$)	0.002 (0.021)	0.002 (0.021)	0.001 (0.021)	0.002 (0.021)
Disaster hits one customer's supplier ($t - 4, t - 1$)	-0.038*** (0.010)			
Disaster hits one customer's specific supplier ($t - 4, t - 1$)	-0.047*** (0.013)	-0.048*** (0.014)	-0.039*** (0.013)	
Disaster hits one customer's nonspecific supplier ($t - 4, t - 1$)	-0.011 (0.013)	-0.013 (0.013)	-0.015 (0.013)	
Observations	139,976	139,976	139,976	139,976
R^2	0.192	0.192	0.192	0.192
Panel C: Controlling for whether any "eventually linked customers'" supplier is hit				
Disaster hits <i>any eventually linked customer's</i> supplier ($t - 4, t - 1$)	-0.008 (0.016)	-0.005 (0.014)	-0.019 (0.012)	-0.012 (0.014)
Disaster hits firm ($t - 4, t - 1$)	-0.039*** (0.013)	-0.040*** (0.013)	-0.038*** (0.013)	-0.039*** (0.013)
Disaster hits one customer ($t - 4, t - 1$)	0.003 (0.021)	0.002 (0.021)	0.004 (0.021)	0.004 (0.021)
Disaster hits one customer's supplier ($t - 4, t - 1$)	-0.032* (0.016)			
Disaster hits one customer's specific supplier ($t - 4, t - 1$)	-0.044*** (0.014)	-0.041*** (0.015)	-0.036*** (0.013)	
Disaster hits one customer's nonspecific supplier ($t - 4, t - 1$)	-0.008 (0.015)	-0.003 (0.014)	-0.008 (0.015)	
Observations	139,976	139,976	139,976	139,976
R^2	0.192	0.192	0.192	0.192
Number of customers' suppliers	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

(b) Table XII: horizontal propagation robustness

TABLE XII
CONTINUED

	Supplier's Sales Growth ($t - 4, t$)			
	Diff.	R&D	Patent	
Year-quarter FE	Yes	Yes	Yes	Yes
Size, age, ROA $\times$ year-quarter FE	Yes	Yes	Yes	Yes
Observations	139,976	139,976	139,976	139,976
R^2	0.192	0.192	0.192	0.192

Notes. This table presents estimated coefficients from panel regressions of firms' sales growth relative to the same quarter in the previous year on one dummy indicating whether one of the firm's customers' other suppliers is hit by a major disaster in the four previous quarters, as well as in Panel A a dummy indicating whether 10% or more of the firm's workforce is hit by a major disaster in the four previous quarters, in Panel B a dummy indicating whether (at least) one location of any other suppliers once in a relationship with a customer of the firm is hit by a major disaster in the four previous quarters. in Panel C

6.10 Figure VI: aggregate input specificity

Read:

- Figure VI uses U.S. input-output data to classify intermediate inputs as exchange-traded, reference-priced, or differentiated.
- The share of differentiated inputs is large and rises over time.
- Differentiated inputs account for more than half of total inputs by the end of the period shown.

Economic message: Since differentiated inputs are common, the supplier-specificity mechanism is likely relevant beyond the paper's disaster setting.

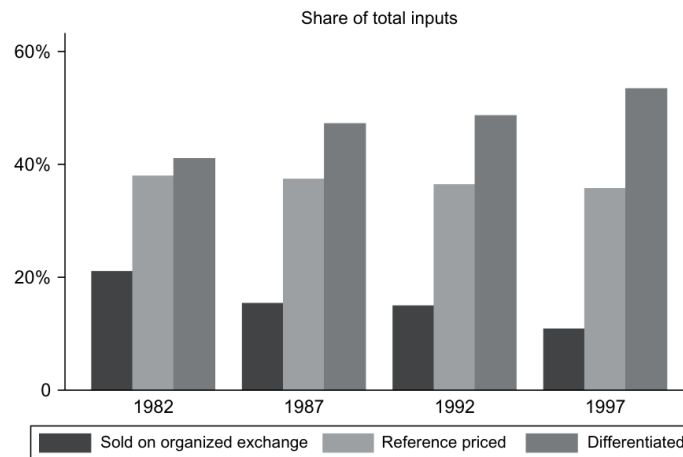


Figure VI

7 Short summary

- The paper asks whether firm-level idiosyncratic shocks propagate through production networks.
- It identifies shocks using major U.S. natural disasters from 1978 to 2013.
- It maps those shocks through Compustat Segment supplier-customer links.
- Directly hit suppliers suffer significant output losses.
- Customers of hit suppliers suffer 2–3 percentage point declines in sales growth.
- Cost of goods sold also falls, consistent with production disruption.
- The effects are stronger when suppliers produce specific inputs.
- Specificity is measured using differentiated goods, R&D intensity, and patenting.
- Customer firms lose market value when their suppliers are hit.
- Shocks also spill over to other suppliers of the same customer.
- The results survive extensive controls for geography, industry, firm characteristics, timing, and placebo relationships.
- The paper provides micro evidence that input specificity is a key mechanism behind shock propagation in production networks.

8 Conclusion

8.1 Core conclusion

Barrot and Sauvagnat show that firm-level idiosyncratic shocks propagate through production networks. Customers experience lower sales and lower market value when their suppliers are hit by major natural disasters. The effect is strongest when the supplier produces specific inputs.

8.2 Economic intuition

The intuition is that firms cannot always replace disrupted suppliers quickly. When inputs are generic, customers can substitute away and the shock is absorbed. When inputs are differentiated, R&D-intensive, or patent-protected, switching is costly. The disruption therefore lowers customer output and spreads through the network.

8.3 Incremental takeaway

The paper’s incremental contribution is to combine a plausibly exogenous firm-level shock, observed supplier-customer links, and direct measures of input specificity. This combination allows the authors to identify both the existence of shock propagation and the microeconomic friction that makes propagation large.

8.4 Broader takeaway

The paper shows why idiosyncratic shocks can matter for aggregate fluctuations. Production networks are not merely accounting linkages; they are systems of input dependencies. When those dependencies involve specific inputs, a local firm-level disruption can produce broad real and financial consequences.